Poppie on Arbitrum: Hub Architecture & Governance Plan

Oct 20, 2026 · Jake Barber

Status: first draft. Contract interfaces are in the Smart Contract Specification.

TL;DR

Poppie lends USDC on Euler V2 against tokenized equities, from about 100 assets at launch to 500.

- **Hubs:** one Prime hub (blue chips only) plus one hub per long-tail risk band. Each band hub holds all blue chips and that band's long-tail names. A band is split only when its hub hits a real limit.
- **Lenders:** deposit USDC into Prime Earn or Yield Earn, which allocate across the hubs.
- **Governance:** all vaults are governed by Poppie, as on BSC.
- **Revenue:** interest fee and rate floor per hub, Earn performance fee, and an enforced origination fee. A slice of Poppie's fee funds a loss reserve.
- **Control:** a risk engine signs changes with KMS keys, and an on-chain steward caps them. Oracle, hook, role and IRM changes go through a 48h timelock.
- **Closed markets:** accounts stay usable. Borrowing uses a haircut price, and only liquidations pause.

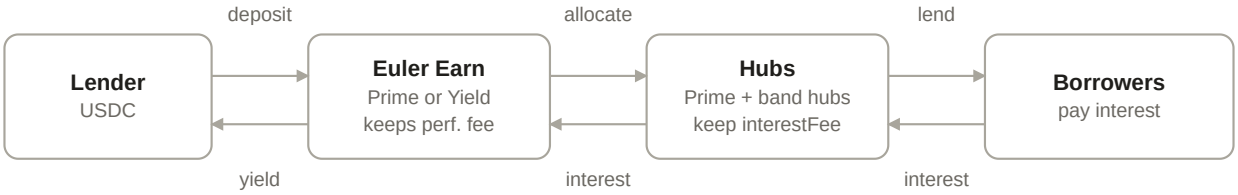
Choice	Gain	Cost
One hub per band	Deep liquidity, simple config, fewer sub-accounts per user	Lenders in a hub share all its names' risk
Governed escrow vaults	Can adjust caps and pause during halts	Users trust Poppie not to block withdrawals
Off-chain exposure rules	Fast and flexible	Limits are soft
Timelock only on admin changes	Risk changes stay fast	Users trust Poppie's off-chain systems for risk changes
Enforced origination fee	A fee on every borrow	Integrators must add one call to each borrow

Euler allows at most 10 collaterals per sub-account, so large portfolios need several sub-accounts. All numbers are proposed starting values.

How it works

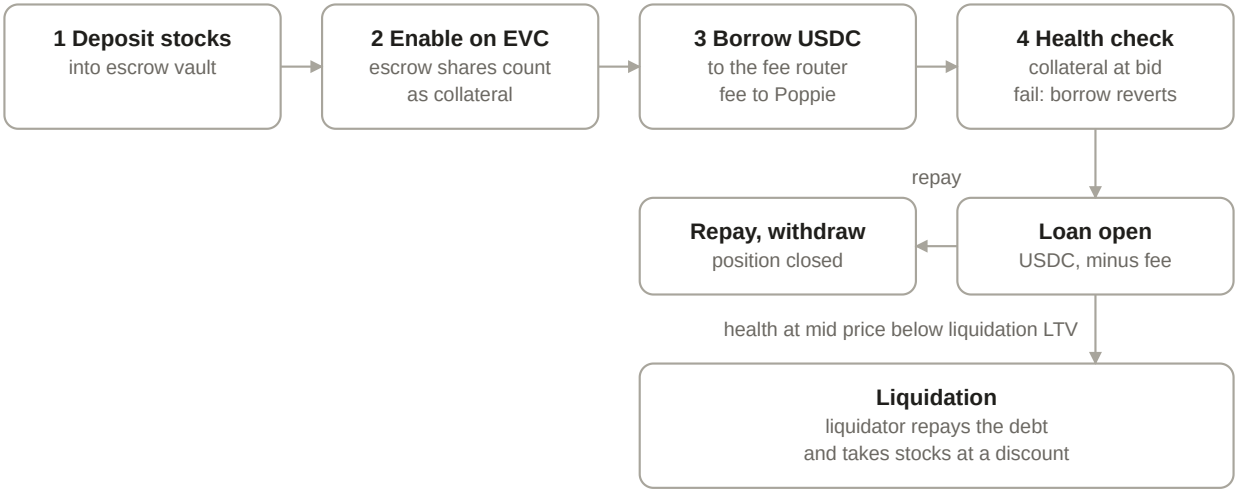
Lenders

Lenders earn borrower interest, minus Poppie fees

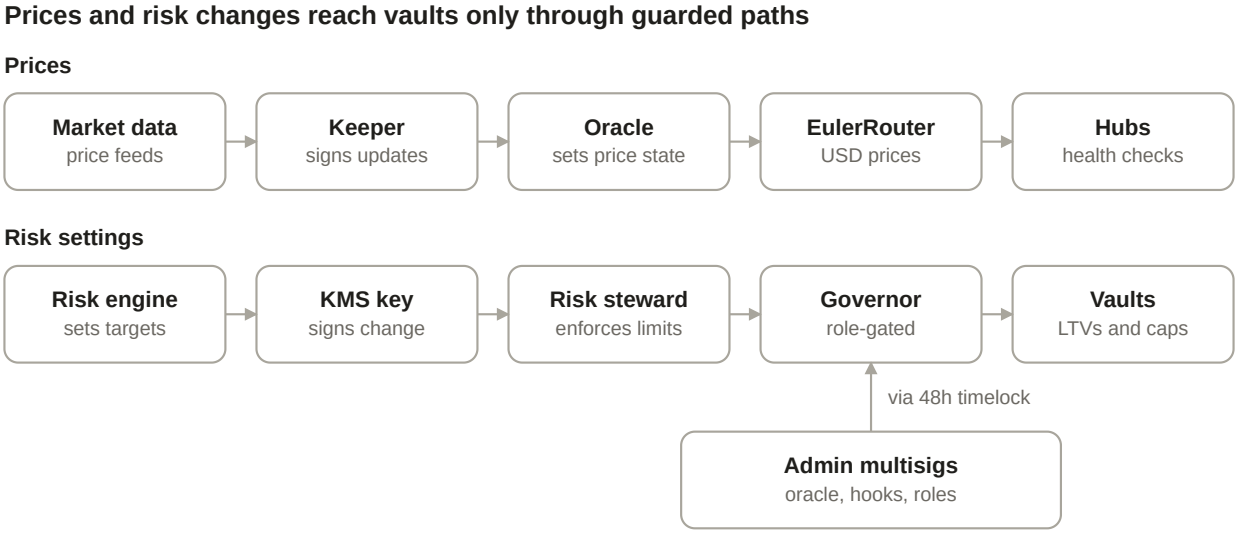


Borrowers

Borrowers post stocks, borrow through the router, and face liquidation



Prices and risk settings



Asset classes

	Blue chip	Long tail
Criteria	Index ETF or top-50 US stock; daily volume ≥ $500M; on-chain exit depth ≥ $1M at ≤ 2% slippage; 90-day volatility ≤ 45%	All other listed names
Max borrow / liquidation LTV	70% / 75%	Set by band (see Hubs)
Max supply cap	20% of exit depth	10% of exit depth
Listed in	Every hub	Its band's hub

Names change class automatically: promoted after 30 days meeting the criteria, demoted after 5 days failing one, with LTV cuts ramped in. A promotion adds the name to the other hubs in one atomic steward call, under launch-style caps, so blue-chip settings can't drift apart between hubs.

Must check first: exit depth on Arbitrum. Caps and liquidations depend on liquidators being able to sell seized stock. If most long-tail Ondo tokens lack on-chain depth, or exit only via Ondo mint/redeem during market hours, the listable set may be far smaller than 500.

Hubs

- **Prime hub:** blue chips only, lowest rates, 8% max liquidation discount.
- **Band hubs (M1–M3):** all blue chips plus every long-tail name in that band. Split a band into two hubs only when one hub hits a real limit (borrow cap, liquidity, or concentration).

Band	Long-tail profile (90-day volatility)	Max borrow / liquidation LTV	Max liquidation discount
M1	≤ 60%, deep liquidity	50% / 58%	10%
M2	60–90%	45% / 52%	15%
M3	> 90%, or thin liquidity	35% / 42%	20%

In each band hub, long-tail-backed debt stays at or below 50% of the hub's borrow cap, and no correlated group exceeds 30% of the hub's long-tail exposure. Both rules are enforced off-chain through supply caps.

	More hubs	Fewer hubs
Gained	Lender risk isolated per hub; settings tuned more finely	Deeper USDC per hub; far fewer LTV entries to sync; most users fit in one hub
Lost	Thin liquidity per hub; ~70 blue chips × N hubs of LTV entries; users split across sub-accounts	Lenders in a hub share every name's risk; one discount and IRM per band

Not affected by hub count: per-asset LTVs and supply caps, the oracle, escrow vaults, the Earn products, and Euler's 10-collateral limit per sub-account. Since Yield Earn spans every hub, extra hubs mainly isolate Prime Earn's lenders, so fewer hubs gives up little.

Fee	Prime	M1	M2	M3
Interest fee	10%	15%	20%	25%
Rate floor	2%	3%	4%	5%
Origination fee	10 bps	15 bps	25 bps	35 bps

Prime Earn puts at least 70% in the Prime hub and charges a 5% performance fee. The 70% is enforced off-chain, because Earn caps are maximums. Yield Earn spans all hubs and charges 10%.

Loss reserve. Hubs have Euler's automatic loss-sharing turned off (`CFG_DONT_SOCIALIZE_DEBT`), so debt left on an account with no collateral stays on that account instead of being spread across depositors. A reserve then repays it with `repay(amount, account)`, which works for any account.

- **Funding:** 10–20% of Poppie's interest fee, a few bps of borrow, is swept into a reserve Safe. It's held in USDC or Prime hub shares, and its balance is published.
- **Payouts:** `K_RESERVE` can only repay hub debt, within per-call and daily limits. Only `MS_OPS` can withdraw from the reserve.
- **Fallback:** if bad debt exceeds the reserve, `MS_OPS` turns loss-sharing back on, and the remainder is spread across the hub's depositors, and through them to Earn depositors.

Why repay quickly: until the reserve repays, the hub's share price overstates its assets, so early withdrawers would exit at full value and later ones would take the loss. The reserve is a buffer for occasional gaps, not insurance against a systemic event.

Origination fee

Euler has no native origination fee, so Poppie enforces one without taking control of user accounts:

1. The user's EVC batch calls `hub.borrow(amount, receiver = feeRouter)`.
2. A borrow hook checks that the receiver is the router and records the amount as pending. It rejects `amount = max uint`, which Euler treats as "borrow all available cash."
3. Later in the batch, the account calls `router.forward(to)`. The router confirms the caller through the EVC, applies the hub's fee rate, and sends `amount − fee` to the receiver the account chose.
4. A hook on the hub's vault status check (`OP_VAULT_STATUS_CHECK`) reverts the batch if any router funds are still pending, so the router never holds funds.

A direct, un-batched borrow always reverts, so integrators must use EVC batches. The status-check hook runs after every operation, including liquidations and repays, and passes when nothing is pending.

Why borrow to the router instead of paying first: the hook runs before the borrow, so a pay-first flow would need USDC up front, which a first-time borrower holding only stock doesn't have.

Why not make the router an EVC operator: an operator can do anything on the account, including withdrawing collateral, so a router bug or upgrade could drain every opted-in user.

Liquidations, repayments and withdrawals are untouched, and the fee is capped at 100 bps on-chain. Integrators add one call to each borrow batch.

Fee basis: every borrow. The fee applies to each borrowed amount, so repaying and re-borrowing pays again. It's the simplest option, has the most precedent (Liquity V1, Aave V1, Abracadabra), and has no bypass paths.

Possible later option: net-new debt only. The router would track each account's peak debt and charge only above it. That's fairer to active traders, but it needs persistent per-account state and peak-reset rules, and must also charge `pullDebt`, so it adds code and audit scope. It has no known precedent in lending.

Flash loans are off at launch. They can be enabled later so liquidators can use hub USDC to repay debts without their own capital, optionally for a fee.

Contracts Poppie deploys on Arbitrum

Contract	Count	Controlled by
PoppieEulerOracle (keeps the last confirmed price through pauses)	1	`MS_ADMIN` via timelock
Three-state price adapter	1	`MS_ADMIN` via timelock
USDC/USD fixed-rate oracle	1	Immutable
EulerRouter	1	`MS_ADMIN` via timelock; `MS_OPS` can only add new routes
IRMs	1 per band, plus Prime	Swapped by `MS_ADMIN` via timelock
Hubs (borrow vaults)	1 + 3	Governor
Escrow vaults	1 per asset	Governor
Euler Earn vaults	2	`MS_OPS`, `K_ALLOC`
Governor, PoppieRiskSteward	1 each	`MS_ADMIN` via timelock
Fee router; hub hook (fee and pause); escrow hook (closed-deposit and pause)	1 each	`MS_ADMIN` via timelock
Timelock (48h)	1	`MS_ADMIN` proposes; `MS_OPS` can cancel
Loss reserve Safe (Roles module, no custom code)	1	`MS_OPS`; `K_RESERVE` can only repay

Why add-only routes: listings stay instant, but a new route can never overwrite an existing price source.
Why IRM swaps are timelocked: a hostile rate model could push borrowers into liquidation.

Euler's core contracts (EVC, factories, lenses) are used at Euler's Arbitrum addresses.

RBAC

Role	Held in	Can	Limit (enforced on-chain)
`K_PRICE`	KMS	Push prices	Moves within the class bound go live; larger moves wait for `K_CONFIRM` or n consistent pushes
`K_CONFIRM`	KMS, separate infrastructure	Confirm unconfirmed moves	Only the latest pending price, while fresh. Off-chain rule: only after its own independent checks pass
`K_TIGHTEN`	KMS	Lower LTVs and caps	Instant; liquidation-LTV cuts ramp over ≥ 72h
`K_LOOSEN`	KMS	Raise LTVs and caps; add promoted names to hubs	≤ 3 pts or +25% per week, up to class ceilings; promotions under launch caps, all hubs in one call
`K_ALLOC`	KMS	Reallocate Earn	Within Earn caps
`K_GUARD`	KMS	Pause deposits and borrows	24h max; never blocks repay or withdraw
`K_RESERVE`	KMS	Repay bad debt from the reserve	Hub `repay` only; per-call and daily limits
`MS_OPS`	Safe 3-of-5	Emergencies, listings (add-only price routes), unpausing, fees, reserve withdrawals, revoking keys	Tighten-only actions are instant
`MS_ADMIN`	Safe 4-of-7	Oracle, adapter, router, hooks, roles, IRM swaps	48h timelock; never a KMS key

If a key leaks:

- `K_PRICE` is trusted. Alone, it can move a live price within the class bound per push, which can trigger liquidations. A larger move goes live without `K_CONFIRM` only after n consistent pushes, and while unconfirmed it can't lift the bid above the last confirmed price × (1 − h).
- `K_ALLOC` can move Yield Earn into the riskiest hubs, up to the Earn caps `MS_OPS` set.
- `K_TIGHTEN`, `K_LOOSEN` and `K_GUARD` can only freeze borrowing or move parameters within the limits above.

Why the 48h timelock on `MS_ADMIN`: an instant oracle swap or a withdrawal-blocking hook could drain or trap funds. These changes are rare, and anything urgent is covered by instant tightening and pauses, so the delay costs almost nothing.

Poppie publishes signer addresses, these limits and every steward action.

Risk automation

The engine watches exit liquidity, volatility, earnings and corporate actions, market sessions, hub concentration and the USDC peg.

When	What runs
Continuously	Exposure checks; tighten automatically on a breach
Hourly	Tightening checks
Daily	Recompute targets; bounded loosening
Before earnings or a corporate action	Lower that name's borrow LTV and cap new supply
Before weekends and holidays	Lower borrow LTVs, mainly on long-tail names

Price states and closed markets

Euler asks the price adapter two questions:

- **Bid** (`getQuotes`): used to value collateral in the account check after every borrow, withdrawal or collateral transfer. The bid sets borrowing power.
- **Mid** (`getQuote`): used for liquidations, including the health check, the discount and the amount seized. The mid decides liquidations.

Poppie's adapter answers the two differently, depending on each name's state.

State	When	Mid → liquidations	Bid → borrowing and withdrawals
Live	Fresh push within the class bound	Latest price	Latest price
Closed	Market closed (nights, weekends, holidays), halt, pause, or no push for 1h	Reverts: liquidations pause	Last price × (1 − h)
Unconfirmed	Fresh price beyond the class bound, e.g. at reopen	Reverts until confirmed (see below)	Lower of the new price and last confirmed price × (1 − h)

The haircut `h` equals the class's reopen bound:

Class	Blue chip	M1	M2	M3
h and reopen bound	10%	15%	20%	30%

Transitions:

- **Closed → Live:** the first fresh push within `h` of the last price.
- **Closed → Unconfirmed:** the first fresh push beyond `h`.
- **Unconfirmed → Live:** whichever comes first:
  - `K_CONFIRM` confirms the latest pending price, at any time after it's pushed, as long as it's still fresh; or
  - the price stays beyond the bound, in the same direction, for n consecutive pushes. A push back within the bound resets the count.

Any state → Closed: no push for 1h, or a pause. The oracle keeps the last confirmed price through pauses, which BSC's oracle doesn't do today.

What the back end must do before confirming. `K_CONFIRM` adds safety only if it checks the price independently. A back end that confirms every large move automatically adds nothing.

- **Detect:** the back end itself flags every move beyond a class bound, as soon as it's pushed.
- **Verify:** an automated check, separate from the pricing path, establishes that the move is real. For example: Poppie's own sources agree, there's a known corporate action or news event, and the conversion multiplier and market session are consistent.
- **Confirm or escalate:** if the checks pass, confirm at once. If they can't decide, page a human immediately; never confirm by default.
- **Record:** log why each move was confirmed or escalated.
- **Fallback:** if nothing confirms, the move goes live after n pushes anyway. A confirmation system that's down delays liquidations by at most n pushes, never indefinitely.

While a name is closed, new deposits of it are blocked. The escrow hook covers deposit, mint, skim and share transfers into an account. Everything else stays open to existing holders.

What this achieves:

- **Accounts stay usable** through weekends and halts. Users can borrow, withdraw and rebalance as long as they're healthy at the bid. Repaying and adding other collateral always work.
- **No forced liquidations.** Neither the haircut nor a wrong price reaches the mid while a name is closed or unconfirmed.
- **No over-borrowing during closures.** Any reopening move within `h` was already priced into the bid. Larger moves wait for confirmation, with the bid at the lower price.
- **No buy-cheap exploit.** Blocking deposits stops anyone buying the token on-chain after bad news and borrowing against the stale bid.

What it costs:

- **Lender risk.** Liquidations wait while a name is closed or unconfirmed.
- **One closed name blocks liquidation of the whole account.** Euler prices every collateral at the mid during liquidation.
- **Two health figures.** "Can borrow" uses the bid and "liquidatable" uses the mid. Interfaces and integrators show both, via `accountLiquidity(account, liquidation)`.
- **A keeper outage looks like a closure.** The behaviour is the same and safe; monitoring tells the two apart.

Listing

1. The engine assigns the name a class, band and hub.
2. One batch deploys its escrow vault, adds its price routes, and sets its LTVs and caps.
3. `MS_OPS` approves instantly, within on-chain constraints:
    - the token is on an approved Ondo token list, which only `MS_ADMIN` can change, via the timelock;
    - new routes may point only to PoppieEulerOracle's adapter and are add-only, both the token's price route and the escrow vault's share-to-asset entry;
    - the escrow vault comes from Euler's factory, verified as escrow;
    - the name starts as long tail, in one hub, under a hard launch cap, at half its class LTV ceiling.
4. The steward steps it up over time.

Why these constraints: without them, an instant listing could add a token Poppie controls, priced as a real stock, and borrow every hub's USDC against it. With them, the worst case is the launch cap × LTV.

Monitoring

Alert when a listed asset has no price route, a hub breaks its exposure rule, a KMS signature doesn't match an engine action, prices go stale, or USDC moves more than 2% off its peg.

- **Keeper liveness:** run redundant keepers with an uptime target, and alert on any missed push. A missed hour sends every name to the closed state.
- **Closed versus outage:** alert when names go stale during market hours, since that means a keeper outage rather than a closure.
- **Monday open:** size liquidator capacity, including backstop liquidity for long-tail names, for the liquidations that resume when names reopen.
- **Two health figures:** interfaces and integrators show both "can borrow" (bid) and "liquidatable" (mid).
- **Bad debt:** alert on any account with debt and no collateral, and repay it from the reserve promptly.
- **Large moves:** alert on every unconfirmed move, and on any that takes longer than the target time to confirm or escalate.

Audits

Track	Scope
Smart contracts	Steward, fee router, hub hook, escrow hook, oracle and three-state adapter (state transitions, bid ≤ mid ≤ ask, every deposit route blocked while closed), role wiring
Off-chain systems	KMS policies, account separation, engine and keeper services, alerting, and whether K_CONFIRM's checks are truly independent of the keeper
Economic	Class and band thresholds, LTV and cap formulas, haircut sizing against real weekend and halt gaps

Rollout

Phase	Scope	Gate to next phase
1 (~100 assets)	Prime hub plus 3 band hubs; both Earn vaults; steward can only tighten	Audits clean
2	Automated loosening	60 days in shadow mode; gas and lens tests with 300+ collaterals in one hub; listing batches reproducible and diffable
3 (→ 500 assets)	More hubs per band	—